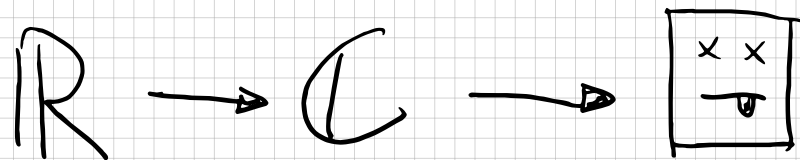


No 3D Complex Numbers



There are no 3-dimensional complex numbers

The complex numbers $\mathbb{C} = \{a+ib \mid a, b \in \mathbb{R}, i^2 = -1\}$ are:

- a two dimensional extension of the reals
- a field — every nonzero element has a multiplicative inverse.
- have no zero divisors — ie if $ab=0$ then $a=0$ or $b=0$.
- multiplication is commutative and associative

claim The only finite dimensional algebras over $\mathbb{R}$ without zero-divisors are $\mathbb{R}, \mathbb{C}$ and $\mathbb{H}$ (associative).

proof: let A be an n -dimensional algebra over $\mathbb{R}$ without zero-divisors. ① A has a multiplicative identity.

Take $0 \neq a \in A$ and $l_a: A \rightarrow A, l_a(x) = ax$ l_a is a lin. transf.

note: $\text{Nul}(l_a) = \{0\}$ if $b \in \text{Nul}(l_a) \Rightarrow l_a(b) = 0$
 $\Rightarrow ab = 0 \Rightarrow b = 0$

$\Rightarrow \exists e \in A$ s.t. $l_a(e) = a \Rightarrow ae = a$.

Sps $b \in A, ab = aeb \Rightarrow a(\underline{b - eb}) = 0 \Rightarrow b = eb$ e is a left identity

Similarly: $r_a: A \rightarrow A, r_a(x) = xa \Rightarrow \exists e' \in A$ s.t. $r_a(e') = a$

$\Rightarrow e'a = a \rightsquigarrow e'$ is a right identity.

$e = ee' = e', \boxed{e=1} \rightsquigarrow \mathbb{R} = \text{span}\{1\} \subseteq A$

② Every non-zero element of A has an inverse.

$$l_a: A \rightarrow A \rightsquigarrow l_a(x) = ax \text{ (surjective)}$$

$$\Rightarrow \exists a' \text{ s.t. } l_a(a') = 1 \Rightarrow aa' = 1 \rightsquigarrow a' = a^{-1}$$

* If $\dim(A) = 1$ then $A \cong \mathbb{R}$ and we are done.

$$\rightsquigarrow \text{sps. } \boxed{\dim(A) \geq 2}$$

$$\text{Take } x \in A \setminus \mathbb{R}, \mathbb{R}(x) = \{a_0 + a_1x + \dots + a_mx^m \mid a_k \in \mathbb{R}\}$$

$\hookrightarrow x$ is not a root of a linear polynomial $\rightsquigarrow x \in A$ *

$$\hookrightarrow x^2 + 2bx + c = 0 \text{ with } b^2 - c < 0 \Rightarrow \dim \mathbb{R}(x) = 2$$

$$\mathbb{R}(x) = \mathbb{R} \oplus \mathbb{R}x$$

$$\text{Set } y = x + b \Rightarrow x = y - b \quad \mathbb{R}(x) = \mathbb{R}(y)$$

$$0 = x^2 + 2bx + c = (y - b)^2 + 2b(y - b) + c = y^2 - b^2 + c$$

$$\Rightarrow y^2 = b^2 - c < 0$$

$$\text{Set } i = \frac{1}{\sqrt{c-b^2}} y \Rightarrow \boxed{i^2 = -1} \quad \mathbb{R}(x) = \mathbb{R}(i) = \mathbb{C}.$$

* If $\dim A = 1$ then $A \cong \mathbb{R}$

$\dim A = 2$ then $A \cong \mathbb{C}$

$$\text{so sps } \boxed{\dim A \geq 3}$$

$$\text{Consider } \sigma: A \rightarrow A, \sigma(x) = -ixi = (i^{-1}xi)$$

$$\overset{\mathbb{R}(i)}{\sigma(xy)} = -ixyi = (-ixi)(-iyi) = \sigma(x)\sigma(y)$$

$$\text{Take } x \in A \setminus \mathbb{C}, \mathbb{C} \subseteq \mathbb{C}(x) \subseteq A$$

claim: $xi \neq ix$

$$\text{proof: sps } xi = ix \Rightarrow (xi)^m = x^m i^m$$

$$\text{Let } p(z) \in \mathbb{C}[z] \text{ satisfied by } x: p(x) = 0$$

$$\parallel \\ (z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_m) \rightsquigarrow x = \alpha_j \in \mathbb{C} \quad *$$

$$\sigma: A \rightarrow A, \sigma(x) = -ixi, \text{ If } x \in A \setminus \mathbb{C} \text{ then } xi \neq ix.$$

$$\text{note: } \mathbb{R}(i) = \mathbb{C} = \{x \in A \mid \sigma(x) = x\}$$

$$\boxed{\text{If } x \in \mathbb{C} \\ \sigma(x) = x \Rightarrow xi = ix}$$

$$\text{Set: } B = \{x \in A \mid \sigma(x) = -x\}$$

$$\text{If } y \in A \rightsquigarrow y = \underbrace{\frac{1}{2}(y - \sigma(y))}_{\substack{\sigma \hookrightarrow \sigma(y) - y \\ \subseteq B \oplus \mathbb{C}}} + \underbrace{\frac{1}{2}(y + \sigma(y))}_{\sigma} \in \mathbb{C}$$

$$\Rightarrow A = B \oplus \mathbb{C}$$

$$\text{So } A = B \oplus \mathbb{C}, B = \{x \in A \mid \sigma(x) = -x\}, \mathbb{C} = \{x \in A \mid \sigma(x) = x\}$$

$$\text{Now take } b \in B \Rightarrow b \in A \setminus \mathbb{R} \rightsquigarrow j \in B, j^2 = -1$$

$$T_j: A \rightarrow A, T_j(x) = j \cdot x$$

$$\text{If } x \in B \Rightarrow \sigma(jx) = \sigma(j)\sigma(x) = (-j)(-x) = jx$$

$$\Rightarrow T_j(x) \in \mathbb{C}$$

$$x \in \mathbb{C} \Rightarrow \sigma(jx) = \sigma(j)\sigma(x) = -jx \Rightarrow T_j(x) \in B$$

$$\left. \begin{array}{l} T_j^4 = \text{id} \end{array} \right\}$$

$$\begin{array}{c} \mathbb{C} \xrightarrow{\tau_j} B \\ \uparrow \tau_j \\ \mathbb{C} \end{array} \Rightarrow \dim B = 2 \Rightarrow \dim A = 4$$

so far: If $\dim A \neq 1, 2$ then $\dim A = 4$ and $1 \in A$ (identity)

$$i \in A, i^2 = -1$$

$$j \in A, j^2 = -1$$

$$\begin{aligned} \text{with } ij \neq ji \quad \sigma(j) = -j &\Rightarrow -iji = -j \\ &\Rightarrow \boxed{ji = -ij} \end{aligned}$$

Next: $k = ij, k^2 = ijij = i(-i)jj = -i^2j^2 = -1$

$$ijk = k^2 = -1$$

$$A = \{ a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}, i^2 = j^2 = k^2 = ijk = -1 \}$$

$$=: \mathbb{H} \quad (\text{Quaternions})$$